Weather and Air Quality Analysis Across Arizona Counties

1. Introduction

This project analyzes how air pollution levels vary with weather conditions across counties in Arizona. The focus is on PM2.5 air pollution and Air Quality Index values, using EPA outdoor air quality data and weather data collected from the OpenWeatherMap API.

The purpose of the project is to build a complete data engineering pipeline that extracts data from multiple sources, transforms and cleans the data, loads it into a relational PostgreSQL database, and then uses SQL queries and visualizations to analyze the results. This project follows the final project requirements by using at least two data sources, including one API source, transforming the data, normalizing it into relational database tables, and producing queries and plots to support the analysis.

2. Project Goals

The project has four main goals:

1. Calculate and compare average AQI levels across counties in Arizona.

2. This goal helps identify geographic differences in air quality across the state.

3. Analyze how temperature relates to air pollution levels.

This goal uses weather data from the OpenWeatherMap API to explore whether counties with higher temperatures also show higher AQI levels.

4. Evaluate how humidity and wind speed influence AQI levels.

This goal examines whether weather conditions such as humidity and wind may be associated with changes in air pollution.

5. Identify the counties with the highest pollution levels.

This goal helps find high-risk counties based on average AQI and PM2.5 values.

3. Data Sources

Source 1: EPA Outdoor Air Quality Data

The first data source was the EPA Outdoor Air Quality Data website. I downloaded daily PM2.5 air quality data for Arizona for 2023.

The EPA dataset included the following types of information:

- Date of observation
- Source
- Site ID
- Daily Mean PM2.5 Concentration
- Units
- Daily AQI Value
- Local Site Name
- County
- State
- Site Latitude
- Site Longitude
- Monitoring method information

This dataset originally contained 9,582 rows and 22 columns, which satisfied the requirement that the data include at least 1,000 rows.

Source 2: OpenWeatherMap API

The second data source was the OpenWeatherMap API. This API was used to collect current weather conditions for representative cities in Arizona counties.

The weather data included:

- City
- Temperature
- Humidity
- Wind speed

The API was important because the project required at least one source collected using an API or web extraction technique. It also provided the weather variables needed to compare air quality with environmental conditions.

4. Data Extraction

The EPA data was extracted by downloading a CSV file from the EPA Outdoor Air Quality Data portal. The selected pollutant was PM2.5, the selected year was 2023, and the selected geographic area was Arizona.

The OpenWeatherMap data was extracted using Python's requests library. For each representative city, an API request was sent to OpenWeatherMap, and the response was returned in JSON format. The JSON response was then converted into a pandas dataframe.

The two datasets were extracted in different formats:

- EPA data: CSV file
- Weather data: JSON API response

This required separate extraction steps before the datasets could be cleaned, transformed, and merged.

5. Data Cleaning and Transformation

<u>EPA Dataset Cleaning</u>

After loading the EPA CSV file into pandas, I first explored the data using:

- head()
- shape
- columns
- info()
- describe()

This confirmed that the dataset had 9,582 rows and 22 columns.

Only the columns relevant to the project goals were kept:

- Date
- County
- State
- Daily AQI Value
- Daily Mean PM2.5 Concentration
- Site Latitude
- Site Longitude

The columns were then renamed to simpler names:

- date
- county

- state
- aqi
- pm25
- latitude
- longitude

The date column was converted from an object/string type into a proper datetime format.

Aggregation to County Level

The EPA dataset originally contained observations at the monitoring site level. This means that multiple rows could exist for the same county because some counties had multiple air quality monitoring sites.

Since the project goals focused on comparing air quality across counties, the data was aggregated to the county level.

First, the data was grouped by county and date:

```
county_daily = epa_clean.groupby(["county", "date"]).agg({
    "aqi": "mean",
    "pm25": "mean"
}).reset_index()
```

Then, the daily county-level values were grouped again to calculate the average AQI and PM2.5 concentration for each county:

```
county_avg = county_daily.groupby("county").agg({
    "aqi": "mean",
    "pm25": "mean"
}).reset_index()
```

This aggregation reduced redundancy and aligned the dataset with the analysis goals. Instead of storing many repeated site-level observations, the final air quality data represented one summarized record per county.

County-to-City Weather Mapping

One important challenge was that the EPA data was organized by county, while the OpenWeatherMap API works best with city names.

To solve this issue, I manually created county-to-city mapping. Each Arizona county was matched with a representative city. For example:

County	Representative City
Maricopa	Phoenix
Pima	Tucson
Coconino	Flagstaff
Yavapai	Prescott
Mohave	Kingman
Yuma	Yuma
Apache	Window Rock
Gila	Globe
La Paz	Parker
Cochise	Sierra Vista
Pinal	Casa Grande

6. Final Integrated Dataset

After cleaning the EPA dataset and collecting weather data from the API, the datasets were merged using the county column.

The final analytical dataset included:

- County
- Average AQI
- Average PM2.5
- Representative city
- Temperature
- Humidity
- Wind speed

The final dataset allowed the project goals to be answered because it combined air quality measures and weather conditions into one unified table.

7. Relational Database Schema

The cleaned and transformed data was loaded into a PostgreSQL database hosted on Aiven.

The database was normalized into three main tables:

Counties Table

The counties table stores county information.

Fields:

- county_id
- county_name

This table acts as a parent table.

Weather Table

The weather table stores weather data collected from the OpenWeatherMap API.

Fields:

- weather_id
- county_id
- city
- temperature
- humidity
- wind_speed

The county_id field connects the weather table to the county table.

Air Quality Table

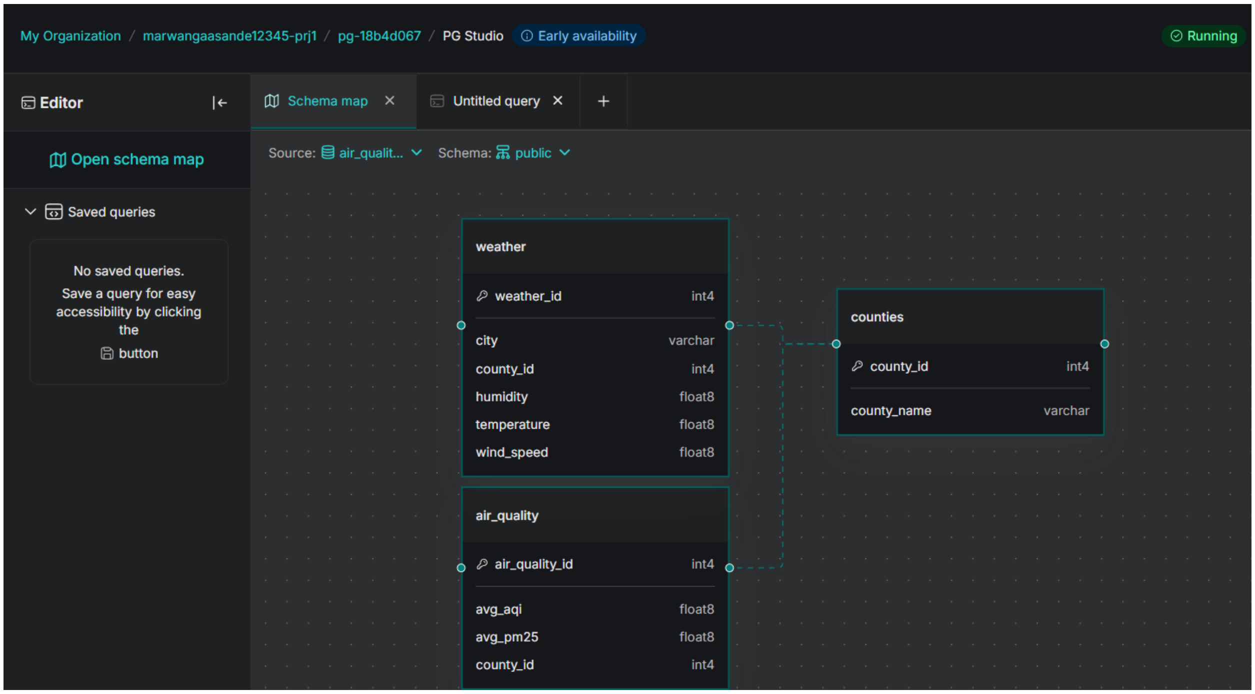
The air_quality table stores summarized EPA air quality data.

Fields:

- air_quality_id
- county_id
- avg_aqi
- avg_pm25

The county_id field connects the air quality table to the counties table.

Schema Diagram



The database design reduces redundancy by storing county names only once in the counties table. Both the weather and air_quality tables reference the counties table through foreign keys.

This structure supports relational queries and makes it easier to join weather and air quality data.

8. SQL Queries and Use Cases

Several SQL queries were used to analyze the data and answer project goals.

Query 1: Full Joined Dataset

This query joins the three tables together to create a complete analytical view.

```
SELECT
   c.county_name,
   a.avg_aqi,
   a.avg_pm25,
   w.city,
   w.temperature,
  w.humidity,
  w.wind_speed
FROM counties c
JOIN air_quality a ON c.county_id = a.county_id
JOIN weather w ON c.county_id = w.county_id;
```

This query confirms that the relationships between the tables are working correctly.

Query 2: Counties with Highest Average AQI

```
SELECT
   c.county_name,
   a.avg_aqi
FROM air_quality a
JOIN counties c ON a.county_id = c.county_id
ORDER BY a.avg_aqi DESC
LIMIT 5;
```

This query identifies the counties with the highest average AQI values.

Query 3: Counties with Lowest Average AQI

```
SELECT
   c.county_name,
   a.avg_aqi
FROM air_quality a
JOIN counties c ON a.county_id = c.county_id
ORDER BY a.avg_aqi ASC
LIMIT 5;
```

This query identifies the counties with the lowest average AQI values.

Query 4: Temperature and AQI

```
SELECT
   c.county_name,
   w.temperature,
   a.avg_aqi
FROM counties c
JOIN weather w ON c.county_id = w.county_id
JOIN air_quality a ON c.county_id = a.county_id
ORDER BY w.temperature DESC;
```

This query supports the goal of exploring the relationship between temperature and air quality.

Query 5: Wind Speed and AQI

```
SELECT
   c.county_name,
   w.wind_speed,
   a.avg_aqi
FROM counties c
JOIN weather w ON c.county_id = w.county_id
JOIN air_quality a ON c.county_id = a.county_id
ORDER BY w.wind_speed DESC;
```

This query supports the analysis of whether wind speed may influence pollution levels.

Query 6: Humidity and AQI

```
SELECT
   c.county_name,
   w.humidity,
   a.avg_aqi
FROM counties c
JOIN weather w ON c.county_id = w.county_id
JOIN air_quality a ON c.county_id = a.county_id
ORDER BY w.humidity DESC;
```

This query supports the analysis of humidity and air quality.

9. Visualizations

Several plots were created from the SQL query results.

The visualizations included:

- Average AQI by county
- Average PM2.5 by county
- Temperature vs AQI
- Humidity vs AQI
- Wind speed vs AQI

The bar charts helped compare pollution levels across counties. The scatter plots examined relationships between weather conditions and AQI values.

These visualizations made it easier to identify counties with higher pollution levels and to observe whether weather conditions appeared to relate to air quality.

10. Findings

Variation in Air Quality Across Counties

The analysis showed that average AQI values varied across Arizona counties. Some counties had noticeably higher average AQI values than others. This suggests that air quality conditions are not uniform across the state.

PM2.5 and AQI Relationship

PM2.5 values aligned with AQI values. This makes sense because PM2.5 is one of the pollutants used to measure air quality. Counties with higher PM2.5 averages tended to have higher AQI values.

Temperature and AQI

The temperature versus AQI plot explored whether warmer areas had higher pollution levels. The relationship was not perfectly linear, but the analysis provided a useful way to compare environmental conditions with air quality.

Humidity and AQI

Humidity varied across the representative cities. However, humidity did not appear to have a strong direct relationship with AQI in this small county-level dataset.

Wind Speed and AQI

Wind speed was also compared with AQI. In general, wind can help disperse pollutants, so this was an important variable to examine. The analysis allowed counties with different wind speeds to be compared against average AQI levels.

11. Challenges

Several challenges came up during the project.

County vs City Mismatch

The EPA dataset was organized by county, but the OpenWeatherMap API required city names. To solve this, representative cities were manually mapped to each county.

This was one of the most important transformation steps because it allowed the two data sources to be integrated.

Site-Level EPA Data

The EPA data was collected at individual monitoring sites. Since the project goal was county-level analysis, the data needed to be aggregated. This reduced redundancy and made the final dataset easier to analyze.

Missing County During Merge

During validation, Pinal County was missing after the merge. This happened because it was not included in the original county-to-city weather mapping. The problem was fixed by adding Casa Grande as the representative city for Pinal County.

API Limitations

The OpenWeatherMap API provided current weather data rather than historical 2023 weather data. Because of this, the weather data was used as a current representative snapshot for each county's selected city. This limitation was documented as part of the project.

12. Data Engineering Summary

This project successfully implemented a full ETL pipeline.

<u>Extract</u>

The data was extracted from two sources:

- EPA Outdoor Air Quality CSV dataset
- OpenWeatherMap API

<u>Transform</u>

The data was cleaned and transformed by:

- Selecting relevant columns
- Renaming columns
- Converting date values
- Aggregating site-level data to county-level data
- Mapping counties to representative cities
- Extracting weather data through API calls
- Merging air quality and weather data

<u>Load</u>

The final cleaned data was loaded into PostgreSQL using a normalized relational schema with three tables:

- counties
- weather
- air_quality

The database was hosted using Aiven PostgreSQL.

<u>Analyze</u>

SQL queries were used to answer the main project's goals, and plots were created to visualize the results.

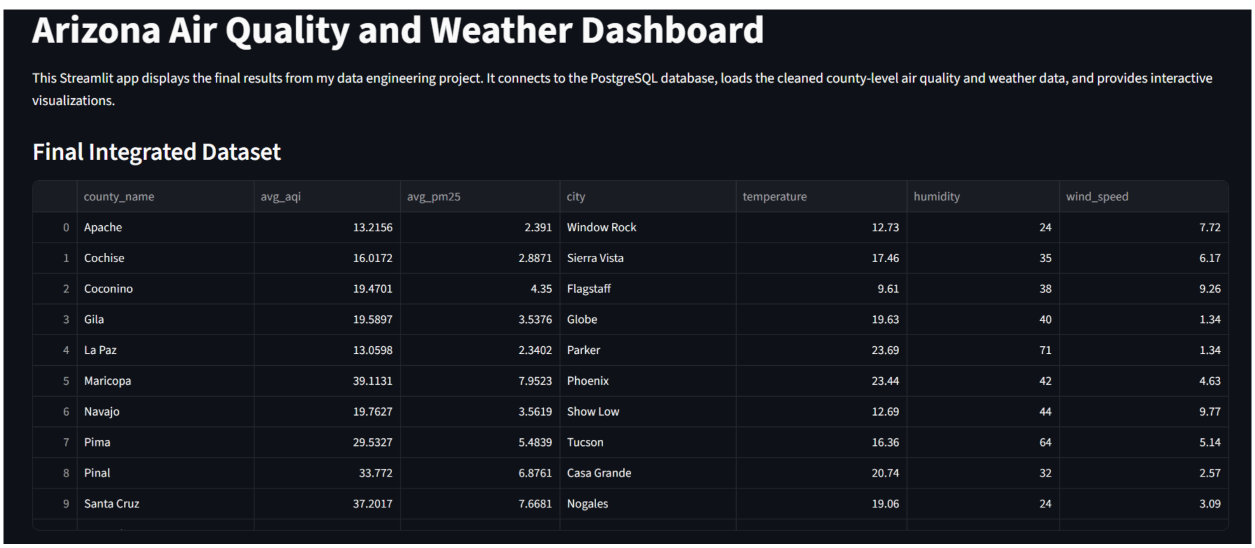
13. Conclusion

This project showed how data from multiple sources can be combined to analyze environmental patterns. The EPA dataset provided air quality information, while the OpenWeatherMap API provided weather conditions. By cleaning, transforming, and merging these datasets, I created a county-level dataset that could be used to study relationships between AQI, PM2.5, temperature, humidity, and wind speed.

The project also demonstrated important data engineering concepts, including extraction from online sources, API collection, data cleaning, aggregation, normalization, relational database design, SQL querying, and visualization.

Overall, the final pipeline successfully transformed raw air quality and weather data into a structured PostgreSQL database that supports useful analysis of air quality patterns across Arizona counties.

EXTRA CREDIT

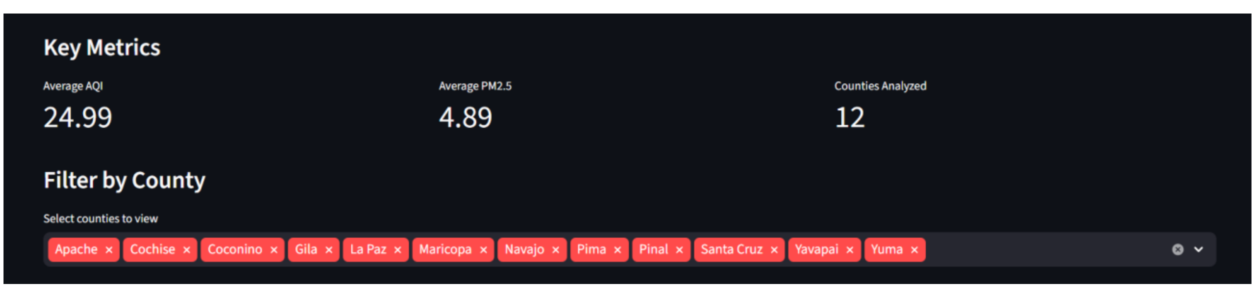


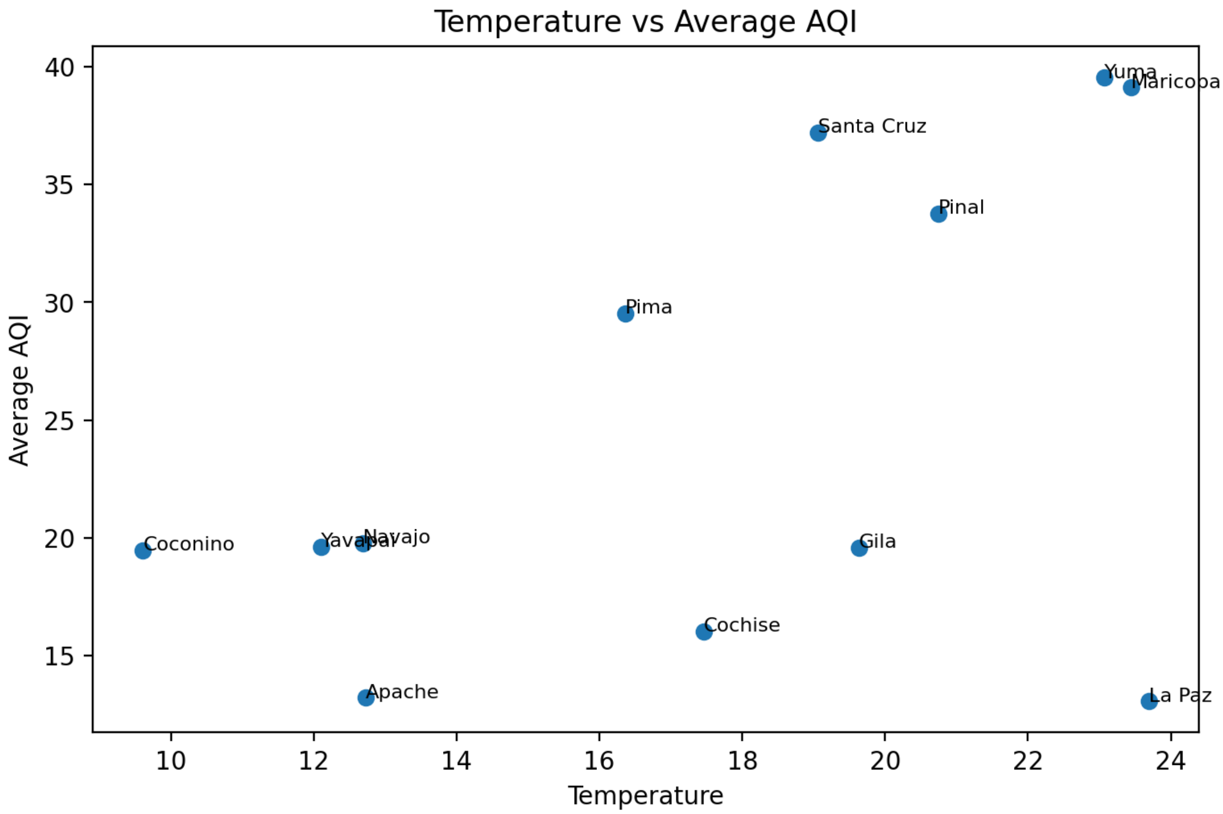
Arizona Air Quality and Weather Dashboard

This Streamlit app displays the final results from my data engineering project. It connects to the PostgreSQL database, loads the cleaned county-level air quality and weather data, and provides interactive visualizations.

Final Integrated Dataset

	county_name	avg_aqi	avg_pm25	city	temperature	humidity	wind_speed
0	Apache	13.2156	2.391	Window Rock	12.73	24	7.72
1	Cochise	16.0172	2.8871	Sierra Vista	17.46	35	6.17
2	Coconino	19.4701	4.35	Flagstaff	9.61	38	9.26
3	Gila	19.5897	3.5376	Globe	19.63	40	1.34
4	La Paz	13.0598	2.3402	Parker	23.69	71	1.34
5	Maricopa	39.1131	7.9523	Phoenix	23.44	42	4.63
6	Navajo	19.7627	3.5619	Show Low	12.69	44	9.77
7	Pima	29.5327	5.4839	Tucson	16.36	64	5.14
8	Pinal	33.772	6.8761	Casa Grande	20.74	32	2.57
9	Santa Cruz	37.2017	7.6681	Nogales	19.06	24	3.09




Temperature vs Average AQI
Average AQI
Temperature
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Pinal
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Yavapai
Navajo
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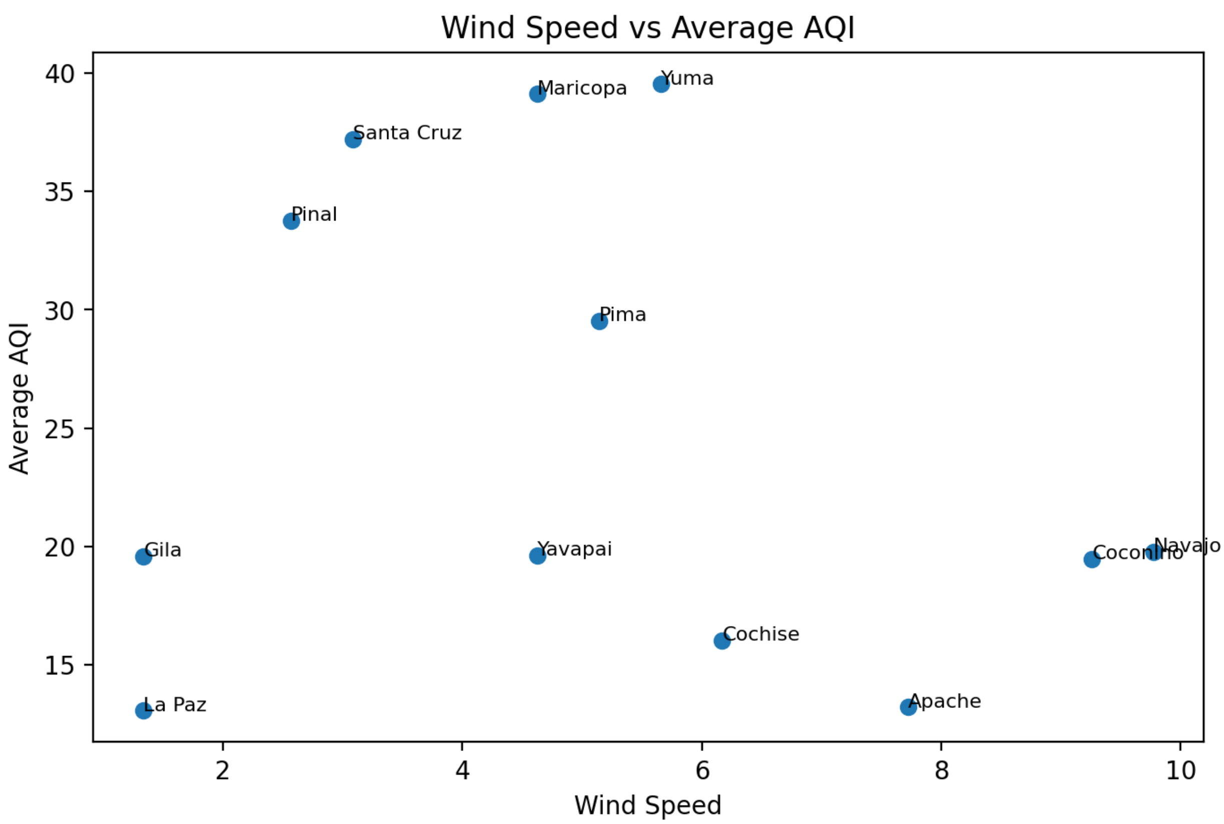
Cochise
Apache
La Paz


Wind Speed vs Average AQI
Average AQI
Wind Speed
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Santa Cruz
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Pima
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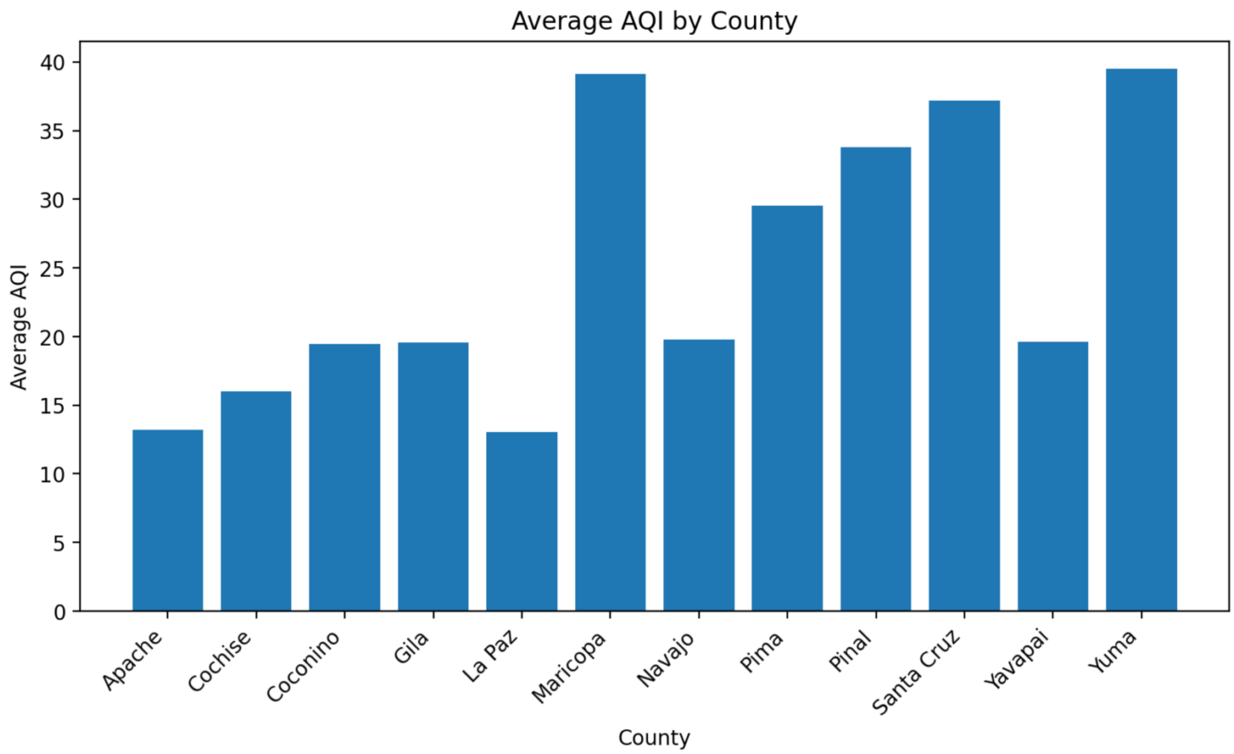

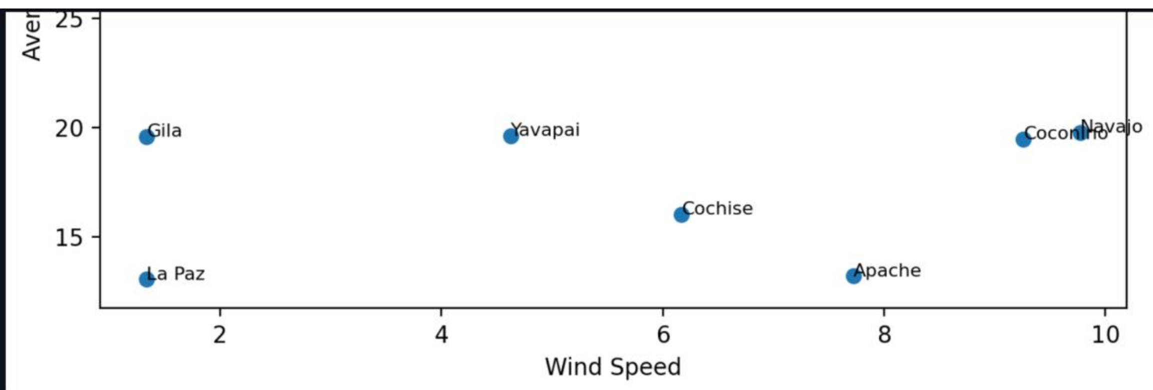

Project Summary

This dashboard shows how air quality varies across Arizona counties and how it relates to weather conditions such as temperature, humidity, and wind speed. The data comes from the EPA air quality dataset and the OpenWeather API, then was cleaned, transformed, and loaded into PostgreSQL.